

522 Assignment 3 rubric

Part 1

1.1 - 10 points

The variables for the top:

- Movie title, nominal, label/text and y-location
- “third” , ordinal/nominal, column/x-location and color
- Rating, quantitative, length

The variables for the bottom:

- Title, nominal, label/text and y-location
- Seen percent, quantitative, length

In the case where there are multiple possibilities, it's ok if they just list one. There should be 3 variables for the top and two for the bottom. Deduct 1 point for each missing variable, and partial (.3) if they have the wrong variable type or wrong encoding.

If they come up with something that seems right but isn't in my list, feel free to reach out.

For the remainder of this rubric, you need to interpret altair 5.5 code.

1.2 - 15 points

We're looking for discussions of part-vs-whole for why pie charts could be good. If we need to accurately measure the difference between movies we might want bar charts. They may also discuss the impact of annotations.

Rubric:

Full credit (15 points): Complete discussion of pros and cons of pie charts in terms of effectiveness. Should have a clear understanding that expressiveness and effectiveness are different. Pie charts can be equally expressive. Recognition that you can use multiple pie charts.

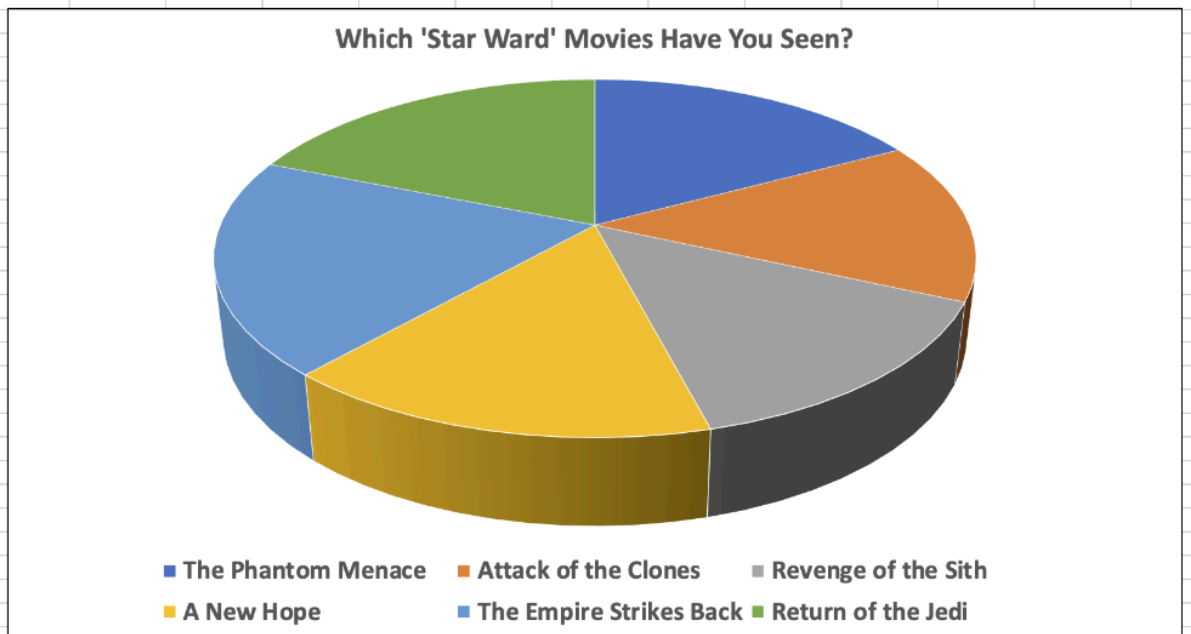
Good (10-15 points): Good explanation of pros and cons. Should have some discussion of parts-vs-whole and acknowledgement of potential benefits. Over focus on pros or over focus on

cons. Over focus on the use of single pie chart to encode the data (closer to 10 points if they don't recognize they can use multiple charts).

Poor (0-10 points): No answer (0) or really shallow ('pie charts are ugly')

If they make mistakes here, point them back to week 2's videos on expressiveness/effectiveness

Examples, this is a poor answer:



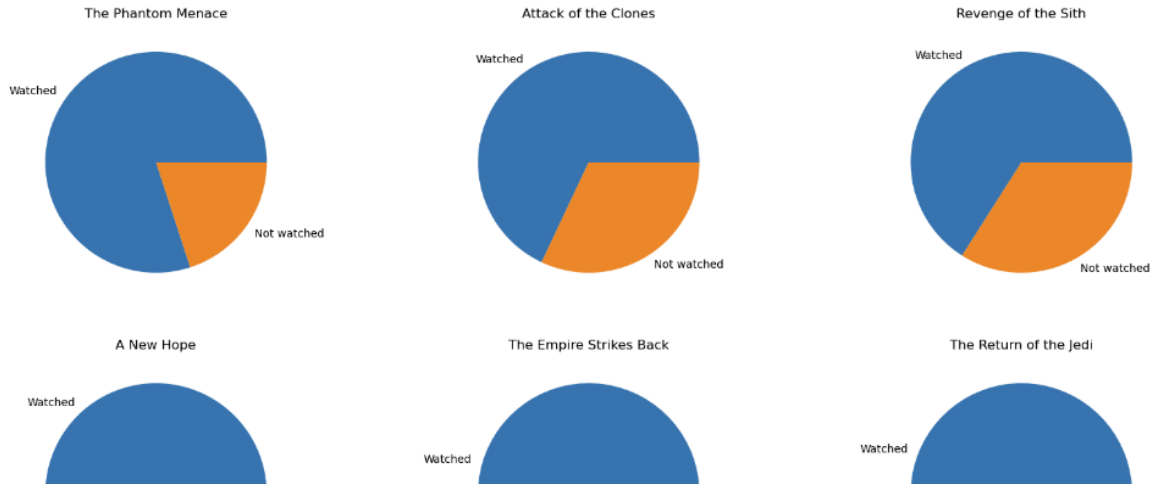
As it can be seeing above, using angles to econde the % of each movie leads to a chart where all movies seem to have the same percentage. The bar chart is much more clear and effective to represent the data.

When the "slices" of the pie are very different (e.g. one variable is 80% and the other 20%), the visualization maybe ok with pie charts. However, the more similar the "slices" are, the more difficult it will be have clarity on what information we are trying to convey.

This is closer to a full credit answer:

It is right when we want to tell what the percentage is like compared with the whole of each movie. Pie chart gives a good illustration on the ratio between the part and the whole (as the figure shows).

It is incorrect when we want to compare the ratio among different movies. Pie chart is not effective compared with bar chart (length). For example, from the pie chart, it is not easy to distinguish which one has higher "watched" percentage between "Attack of the Clones" or "Revenge of the Sith", but we can directly tell it from the bar chart by simply comparing the length of the bars.

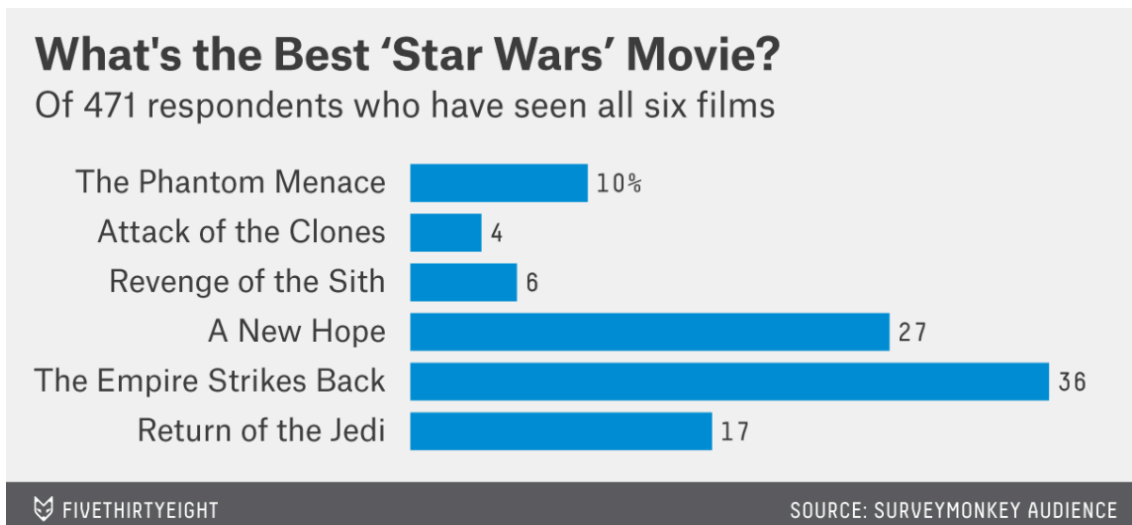


(If they added percent labels onto the pie, the ratios might be easier to calculate, but that's a quibble, I'm not sure I'd deduct points for that).

Part 2

2.1 - First vis - 10 points

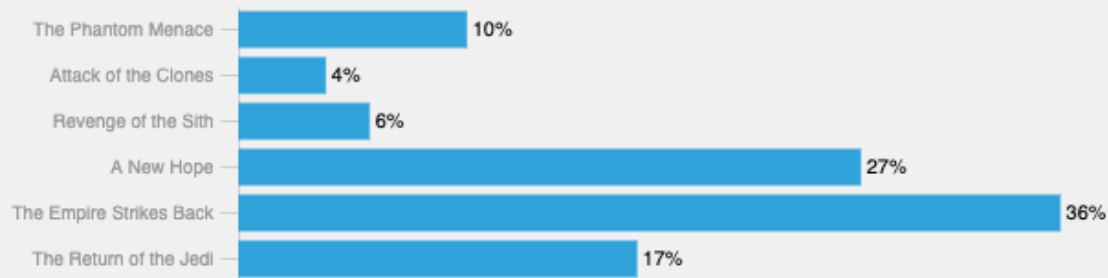
538 solution:



Acceptable solution:

What's the best 'Star Wars' Movie?

Of 471 respondents who have seen all six films



They at least need the “acceptable solution” but if they get closer to the 538 that’s great
Things to look for:

- Right style/color/font/title/subtitle/background color (-1 for each wrong thing, up to -3)
- Movies in the right order (-2 if not)
- Percents should match (-5 if really off, -1 if a little off)
 - Should have 471 respondents, percents down the list: 10, 4, 6, 27, 36, 17

If they’re not matching on style, point them to:

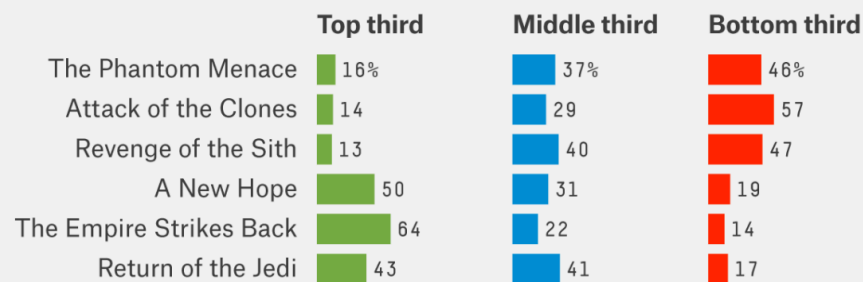
`alt.themes.enable('fivethirtyeight')`

2.2 Second vis - 10 points

538:

How People Rate the ‘Star Wars’ Movies

How often each film was rated in the top, middle and bottom third
(by 471 respondents who have seen all six films)



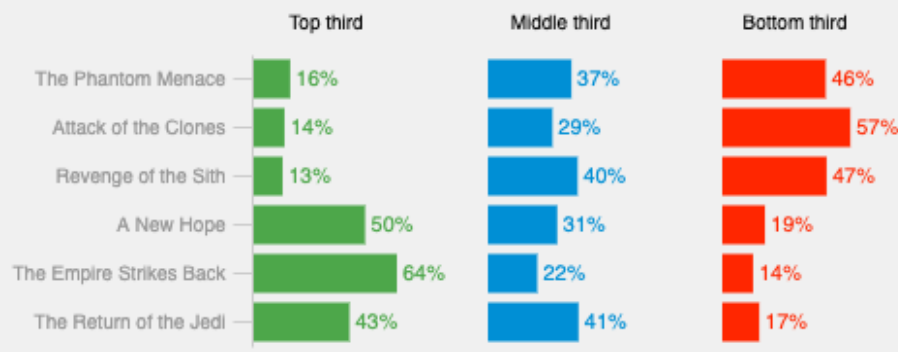
FIVETHIRTYEIGHT

SOURCE: SURVEYMONKEY AUDIENCE

Acceptable:

How People Rate the 'Star Wars' Movies

How often each film was rated in the top, middle and bottom third
(by 471 respondents who have seen all six films)



Things to look for:

- Right style/color/font/title/subtitle/background color (-1 for each wrong thing, up to -3)
- Movies in the right order (-2 if not)
- Percents should match (-5 if really off, -1 if a little off)
 - Should have 471 respondents, spot check percents
 - Bars should be scaled correctly! Bottom green and bottom blue bars should be around the same length (and just slightly smaller than top red)
 - If this is off, they might not be faceting correctly, you can point this out as an option
- If they concatenated charts and didn't facet, point them to faceting as an option, e.g., "You will likely get a better result for this exercise by using Altair's facet features."

If they're not matching on style, point them to :

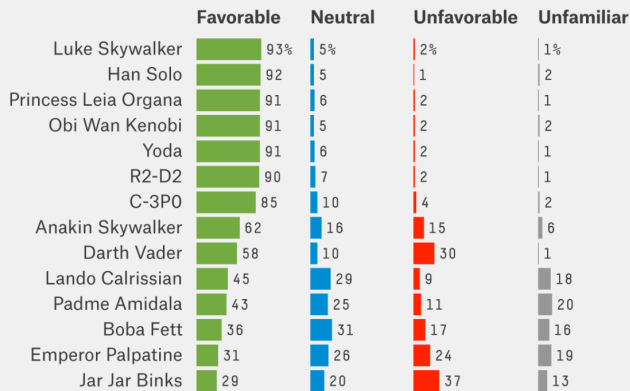
`alt.themes.enable('fivethirtyeight')`

2.3 Third vis - 10 points

538:

'Star Wars' Character Favorability Ratings

By 834 respondents



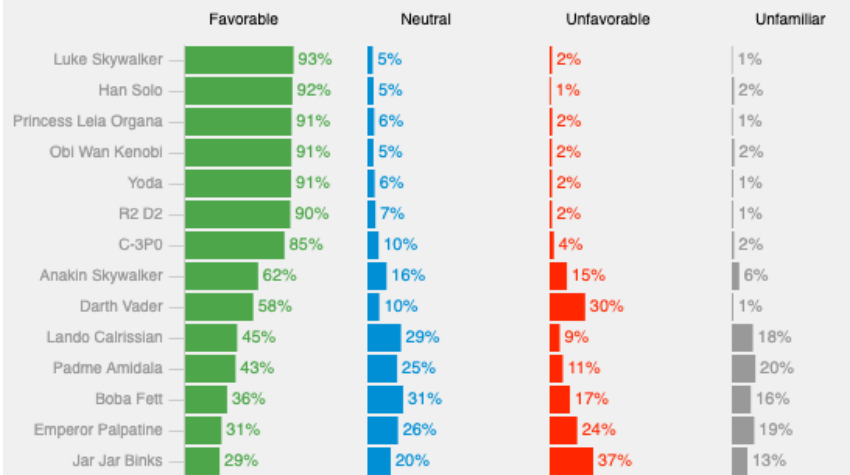
FIVETHIRTYEIGHT

SOURCE: SURVEYMONKEY AUDIENCE

Acceptable:

'Star Wars' Characters Favorability Ratings

By 826 respondents



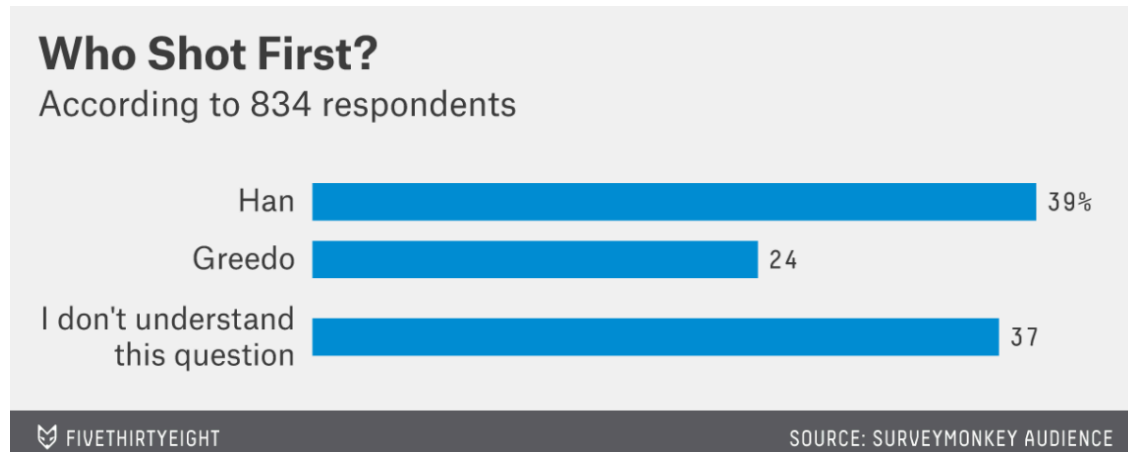
Things to look for:

- Right style/color/font/title/subtitle/background color (-1 for each wrong thing, up to -3)
- Characters in the right order (-2 if not)
- Percents should match (-5 if really off, -1 if a little off)
 - Should be 826 or 834 respondents, spot check percents,
 - Bars should be scaled correctly! Bottom green and bottom blue bars should be around the same length (and just slightly smaller than top red)
 - If this is off, they might not be faceting correctly, you can point this out as an option
- If they concatenated charts and didn't facet, point them to faceting as an option, e.g., "You will likely get a better result for this exercise by using Altair's facet features."

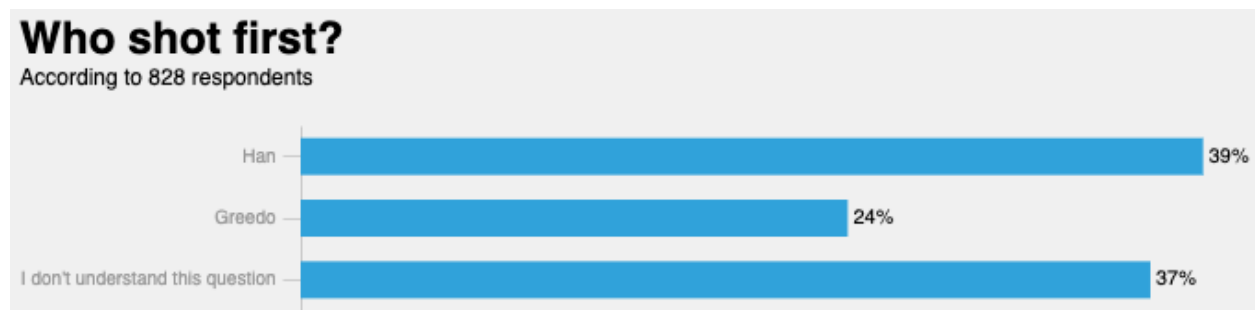
If they're not matching on style, point them to :
`alt.themes.enable('fivethirtyeight')`

2.4 4th vis - 10 points

538:



Acceptable:



They at least need the “acceptable solution” but if they get closer to the 538 that’s great

Things to look for:

- Right style/color/font/title/subtitle/background color (-1 for each wrong thing, up to -3)
- characters in the right order (-2 if not)
- Percents should match (-5 if really off, -1 if a little off)
 - Should have 834 or 828 respondents, percents down the list: 39, 24, 37

If they're not matching on style, point them to :

Using Altair themes will help with styling details. Please see `alt.themes.enable('fivethirtyeight')`

Part 2 Hard coding test

Double check for hard coding of the dataset. If we had provided a different dataset in the input argument would the student's function have skipped it?. The visualizations generated use less

data and should look like the following. -5 if their functions are hard coded and generate the wrong data distributions (once). If the only thing they mess up is the #'s in the subtitle—probably the most common mistake—just do -2 (once). Part 3 - Make your own
We're looking for two visualizations. The implementation of which should be 20 points.

One should be obviously better than the other. The two visualizations should use different marks and at least some of the encodings should be different. Visualizations do not need to be different in terms of expressiveness but should be different in effectiveness.

Implementation Rubric:

- Full credit (20) points: good faith effort on both visualizations, visualizations needs to have something to do with the article. Visualizations use different mark types (as specified) and at least some different encodings.
- Partial credit (10-20) points: good faith effort on both visualizations. Vis has something to do with article. Visualizations may not follow spec: use the same mark type or same encoding.
- Low credit (0-10 points): no implementation or missing one visualization, visualizations have 0 to do with article, no obvious differences,

The justification should make clear what the expressiveness/effectiveness differences are. Partial credit is fine here based on how much they missed.

Justification rubric:

- Full credit (15 points): completely and accurately describe the effectiveness and expressiveness differences between the two visualizations. Correctly understand effectiveness and expressiveness (i.e., they don't make an expressiveness argument based on effectiveness).
- Good (10-15 points): Are largely correct in distinguishing between the two visualizations. May have missed one or two points. Correctly understand effectiveness and expressiveness (i.e., they don't make an expressiveness argument based on effectiveness).
- Fair (5-10 points): Made significant mistakes or omissions in contrasting the two visualizations. Do not seem to correctly understand expressiveness/effectiveness.
- Poor (0-5 points): Missing or extremely poor argument, no discussion of expressiveness/effectiveness. Shallow discussion of differences. Complete misunderstanding of differences.